

Petrology of anorthosites from the Descartes region of the moon: Apollo 16

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Abstract—Seven large and six 2–4-mm size anorthosites from the Apollo 16 site have been studied by petrographic and microprobe techniques. These rocks appear to be remnants of the primitive cumulate crust (10–40 km) that were excavated either by the Nectaris impact and thrown into the Apollo 16 site or alternatively were derived more locally and were excavated by “Unknown Crater A” (Head, 1975). Before excavation the anorthosites experienced a long period of annealing and subsolidus crystallization. Augite and orthopyroxene pairs indicate thermal equilibration to temperatures less than 800°C. Coexisting opx and feldspar show a positive correlation between albite contents of the feldspar and Fe/Fe + Mg of the trapped opx which is the normal trend to be expected in igneous differentiation. If the feldspar cumulates floated, one would expect the more albitic feldspars and iron-rich trapped pyroxenes to occur at greater depths. Our samples and those of others definitely show a bias toward more magnesium-rich pyroxene composition which is consistent with the above hypothesis in that these rocks would be nearer the surface and thus more readily excavated by medium-sized impacts.

INTRODUCTION

THE APOLLO 16 MISSION to the Descartes region of the moon provided us with perhaps the best sampling of highland rock types. In a previous study by Delano *et al.* (1973), it was found that 9% of 342 fragments in our Apollo 16 2–4-mm collection were anorthosites very similar to large anorthosites such as 15415 from Apollo 15. Anorthosite is considered by many to have accumulated during the original lunar differentiation and to have preserved in part its cumulate texture. Because of their apparent importance in lunar petrologic history, the anorthosites have received considerable attention by the scientific community. The “genesis rock” 15415 has been petrologically characterized by Steele and Smith (1971), James (1972), Hargraves and Hollister (1972), and Stewart *et al.* (1972). Large anorthosite 60015 has been studied by Sclar *et al.* (1973) and Sclar and Bauer (1974), and 60025 has been studied by Hodges and Kushiro (1973). Meyer and McCallister (1973) examined rock 60215, and anorthositic breccia 67075 has been extensively studied by Brown *et al.* (1973), Peckett and Brown (1973), and McCallum *et al.* (1974). Steele and Smith (1973) described 61016 and 67075. Dowty *et al.* (1974) reported on anorthositic rake samples from Apollo 16. The present study is intended to complement these and other excellent studies. Our objectives include interpretation of the thermal history of these rocks, definition of the

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petrogenetic interrelationships and the definition of a possible stratigraphic sequence for the Apollo 16 anorthosites.

The specific samples studied included large rocks 60015 sections 113, 122, and 123; 60025 sections 108, 110, 117, 120, 128, and 129; 60215,13; 61016,217; 64819,3; 65315,89; and 67075,48. In addition, six fragments from the 2–4-mm size range were examined: 63503,13 sections 12 and 21; 64503,1 sections 26 and 36; 66043,1,4; and 68503,16,22 (see Table 1). We are particularly interested in anorthosites that are monomict in nature.

“Anorthosite” is a descriptive name generally interpreted by lunar scientists as a “lunar rock with over 90% modal or normative plagioclase” (Glossary of Lunar Terms, Proc. Lunar Sci. Conf. 5th, Vol. 1). We will use a more genetic definition: a rock with >95% modal feldspar which shows no evidence suggesting a polymict nature, i.e. uniform mineral compositions and monomict petrographic texture. Other anorthositic rocks, described as anorthositic breccias, contain mineralic and lithic clasts which originated as true anorthosites.

The anorthosites included in this study display several properties in common. All samples contain <2% pyroxene which in most cases has equilibrated to an augite–hypersthene pair. They also display similar petrographic textures indicative of shock from a hypothetical excavation event.

Investigations were carried out using petrographic and chemical analytical techniques. Electron microanalyses were performed with a computer driven, 4-spectrometer ARL-EMX SM microprobe with real-time data reduction. Matrix correction is according to Bence and Albee (1968) with modification of the matrix by Albee and Ray (1970). Glasses and feldspar were usually analyzed with a 10–20- μ beam, while pyroxene was done with a finer beam ($\approx 5 \mu$). Al, Ca, Na, and K standards were feldspars; other elements were standardized on pyroxenes and oxides.

Table 1. Samples included in this study.

	Textural type	Pyroxene K_D	Mass of collected sample (gm) ¹	Location
60015	II	0.612	5574.0	Sta 10—ALSEP
60025	I	0.616	1836.0	Sta 10—ALSEP
60215,13	III	0.528	385.8	Sta 10—ALSEP
61016,217	II	0.526	11729.	Sta 1
63503,13,12	II	0.59]	14.53	Sta 13
63503,13,21	II	0.65]		
64503,1,26	III	0.59]	24.11	Sta 4, Trench bottom
64503,1,36	II	0.72]		
64819,3	II	0.532	11.76	Sta 4, Crater rim
65315,89	III	0.567	300.4	Sta 5
66043,1,4	III	0.95	15.5	Sta 6
67075,48	III	~ 0.64	219.2	Sta 11
68503,16,22	Recrystallized	—	25.10	Sta 8

¹Weights from Apollo 16 Lunar Sample Information Catalog (1972).

PETROGRAPHY

Apollo 16 anorthosites display a range of texture, grain size, and shock damage. In spite of these textural variations the mineral chemistry remains remarkably intact. Most textural differences may be directly related to shock damage caused by crater excavation and subsequent surface impact history. The initial excavation from deep in the lunar crust may be largely responsible for the cataclastic texture of the anorthosites. After excavation some of the anorthosites witnessed brecciation and/or cataclasis from much smaller scale surface impacts. The range of textures displayed may be conveniently divided into three types. Type I represents the least-damaged texture while Types II and III represent more intense shock damage and breccia formation, respectively.

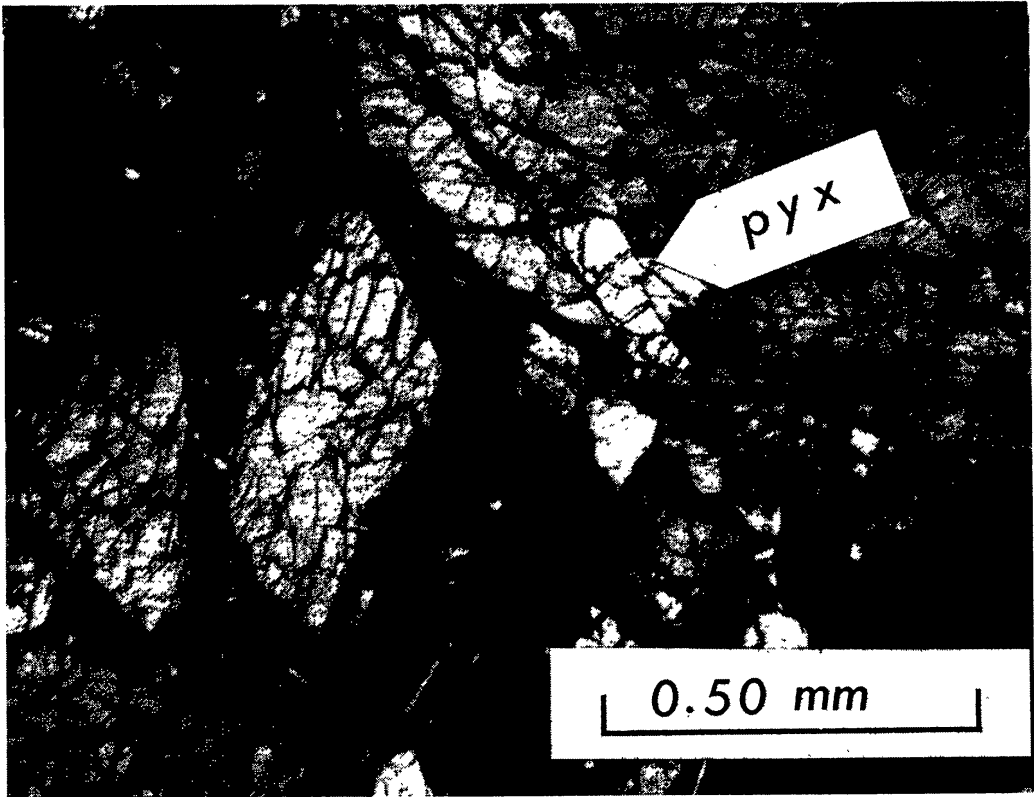
Type I

Type I textures are typified by anorthosites 60025 (Figs. 1a and 1b) and 15415. Sample 60215,13 (Fig. 1c) is a polymict breccia but much of its texture is *similar* to Type I. 60025 sections 108, 110, 117, 120, 128, and 129 display uniform petrographic features. Hodges and Kushiro (1973) and Walker *et al.* (1973) have previously described the petrographic nature of 60025. Modal mineralogy (see Table 2) is dominated by greater than 98% anorthite feldspar. Large anhedral, slightly rectangular feldspar (with grain size averaging 1–2 mm and up to 10 mm) is surrounded by a matrix of intermediate size and granulated feldspar. The intermediate grains are polygonal, display infrequent twinning, extinct uniformly, and average between 0.1 and 0.5 mm. The granulated feldspar is often annealed, polygonal, and shows uniform optical extinction.

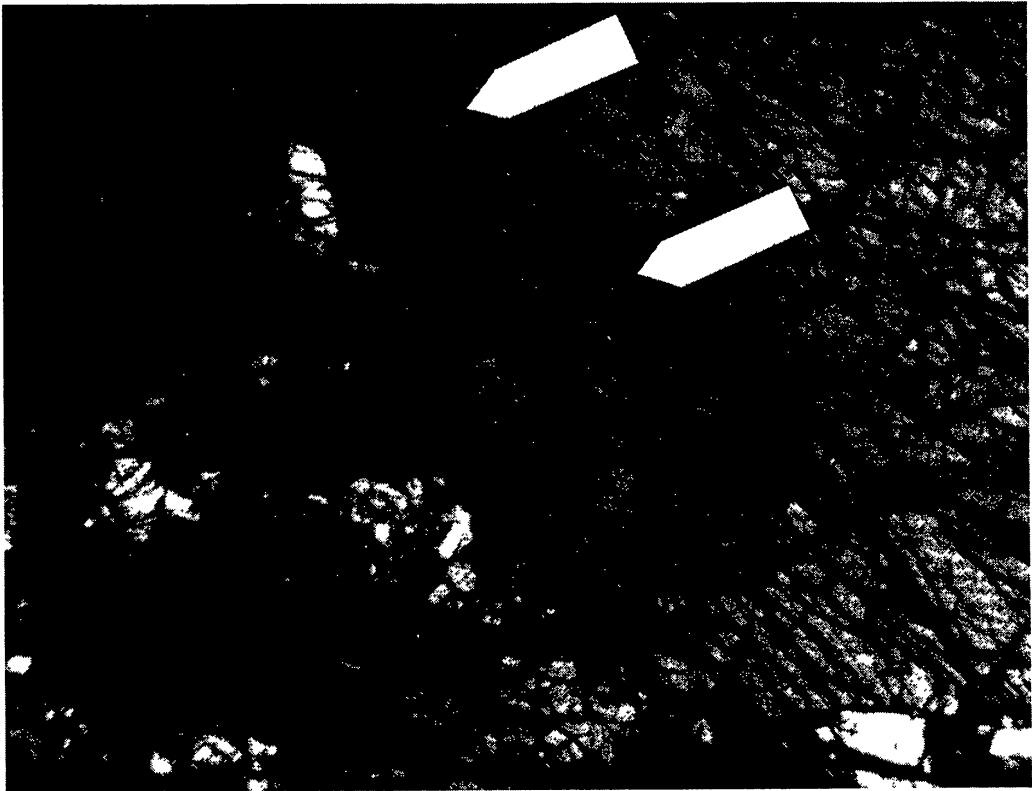
The largest feldspar grains display a variety of cataclastic shock features. Shock-induced twinning is prominent and most intense near large fractures. Twin lamellae often include glassy blebs, and other instances of local melting are numerous. Extinction is usually undulatory or patchy, and fractures offset most linear features (i.e. twinning and grain boundaries). Such fractures are often glass filled and may be limited to a single grain or may cut the entire section.

Pyroxene usually occurs as anhedral grains (generally < 0.5 mm) usually in the granulated matrix, but some occurs as long, thin stringers along feldspar twin lamellae. Smith and Steele (1974) have interpreted these pyroxene stringers as exsolution products from feldspar. We discuss this problem below. Orthopyroxene also occurs as inclusions within the feldspar, and these included pyroxenes are usually found near grain boundaries (Fig. 1a).

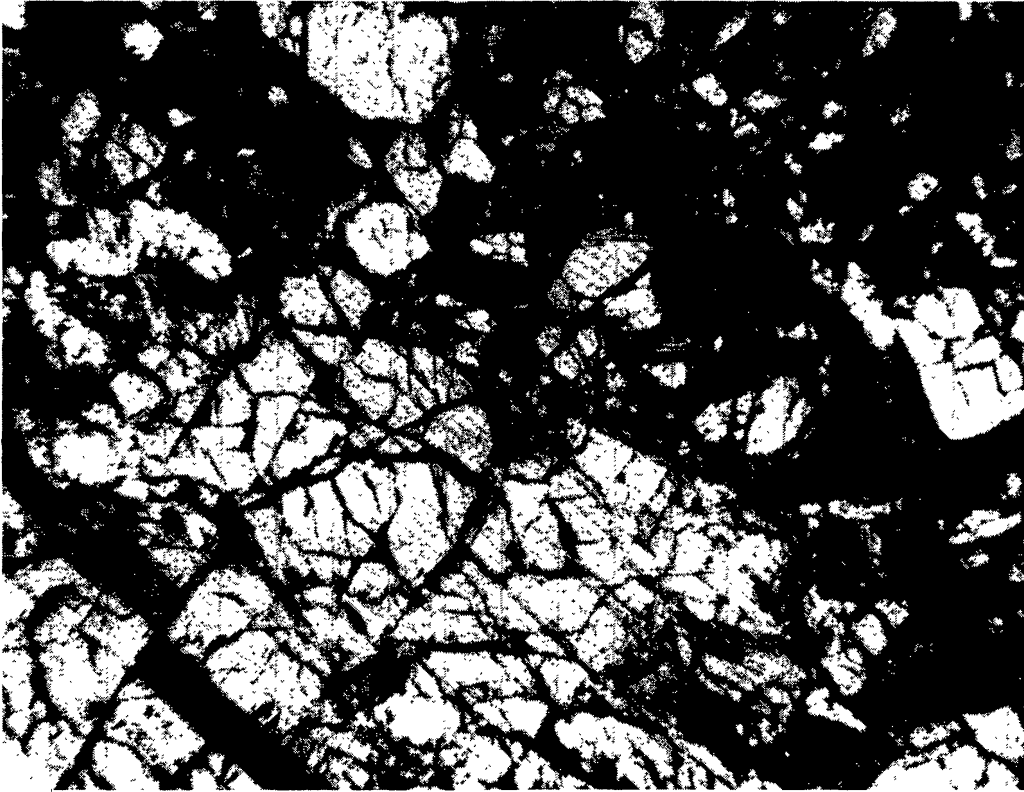
Pyroxene is not abundant in these anorthosites and usually occurs in areas of aggregates. The aggregates include both ortho- and clinopyroxenes as well as minor amounts of ilmenite and chrome-rich spinel. Orthopyroxene is dominant (~90% of all pyroxene) and usually forms discrete grains. Augite occurs as epitaxial grains and as narrow exsolution lamellae in opx. Some augite occurs as single discrete grains and is recognized in thin section by its higher birefringence. No olivine was found in any of the six sections studied here, nor was any reported by Hodges and Kushiro (1973).



(a)



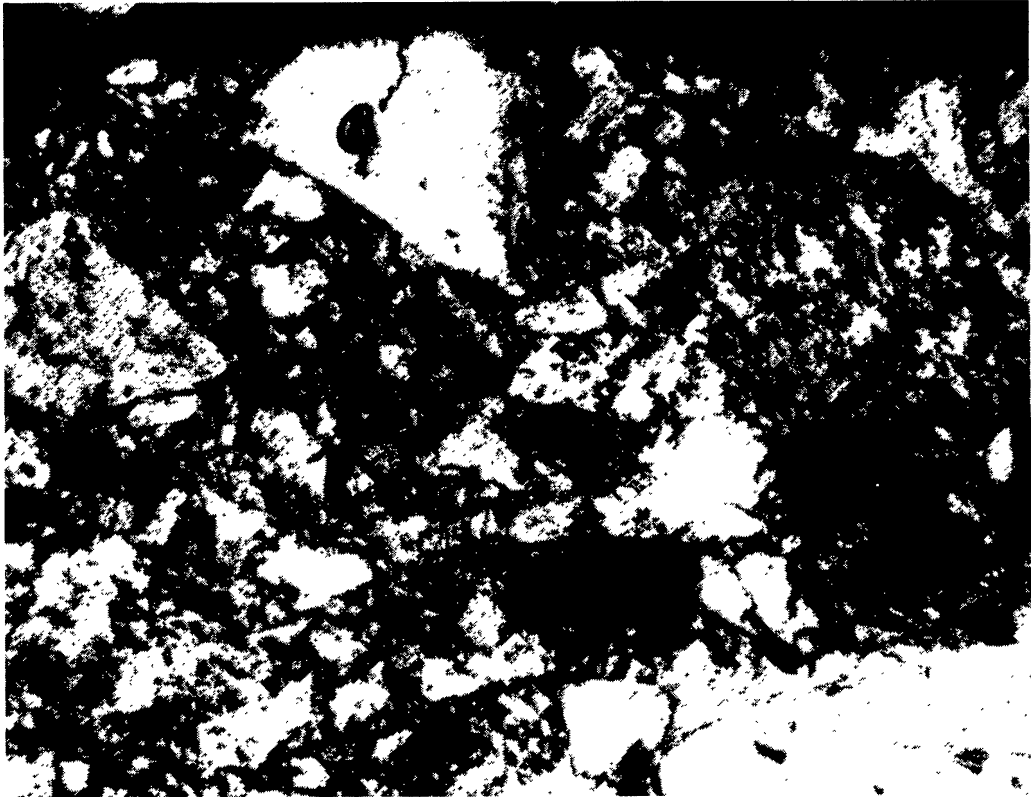
(b)



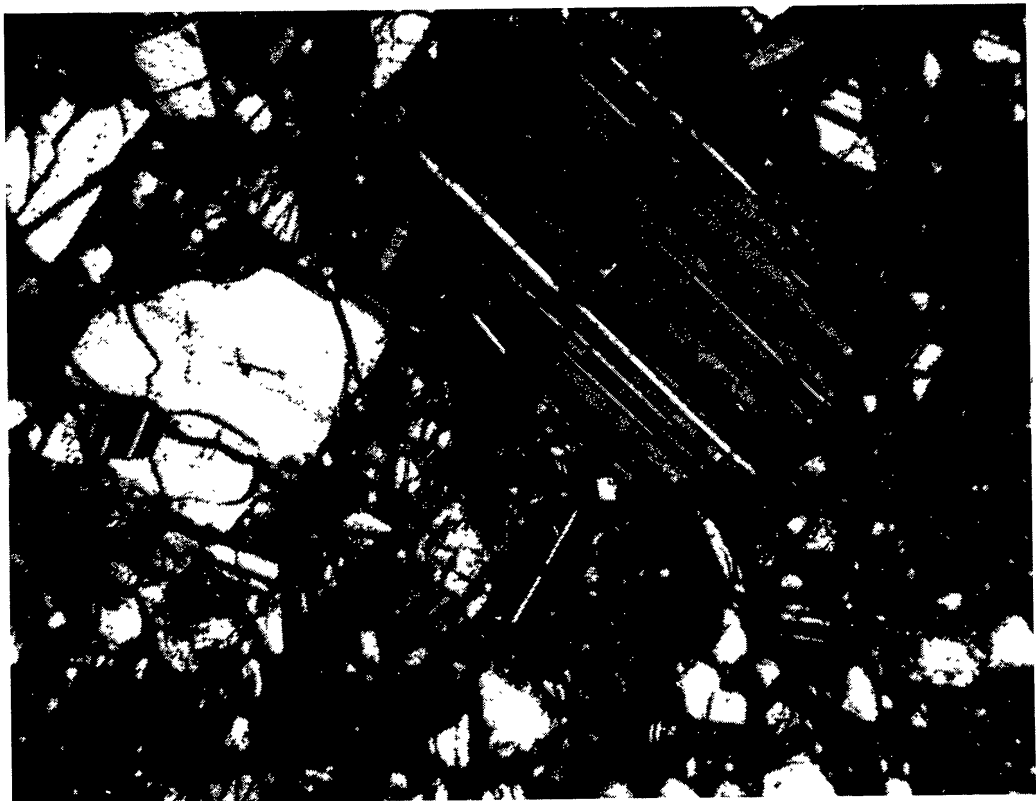
(c)



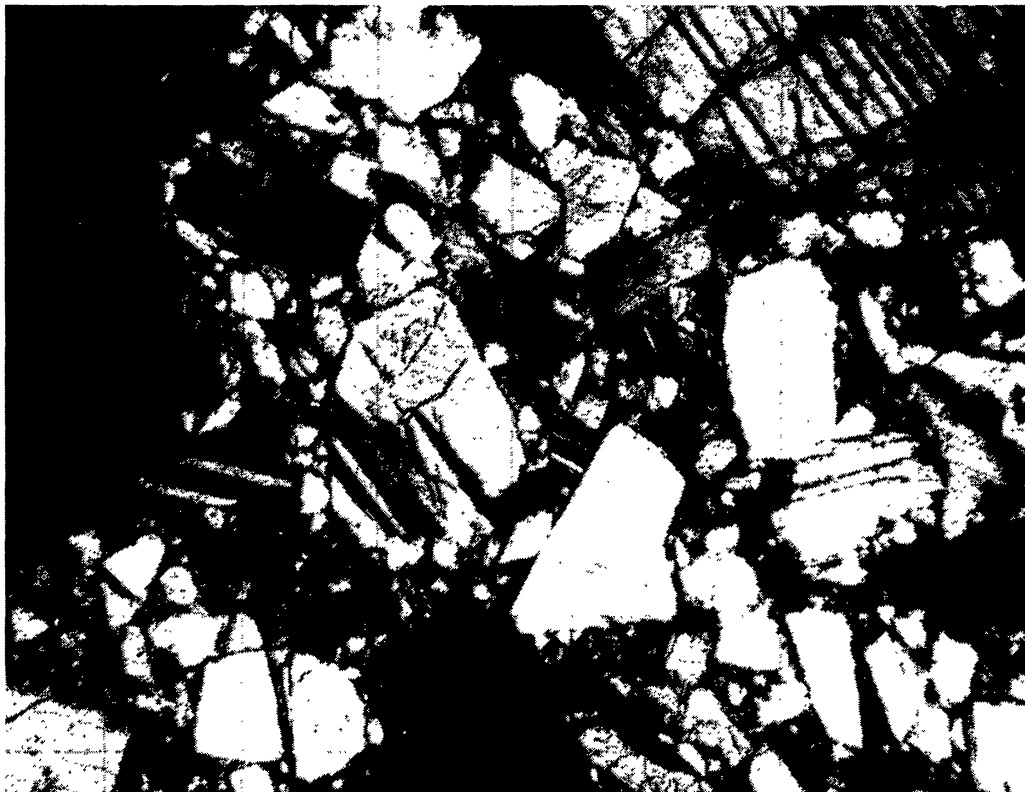
(d)



(e)



(f)



(g)

Fig. 1. (a) 60025,120: large feldspar grains surrounded by finer-grained matrix. Arrow points to included orthopyroxene with NE-SW trending augite lamellae. Crossed polars. Long dimension = 1.4 mm. (b) 60025,120: large feldspar grain. Fractures offset albite twin (arrows). Crossed polars. Long dimension = 1.4 mm. (c) 60215,13: large, damaged feldspar with matrix. Crossed polars. Long dimension = 1.4 mm. (d) 60015,113: triple junction of feldspar occupied by pyroxene. Crossed polars. Long dimension = 1.4 mm. (e) 64819,3: crushed, "feathery" matrix. Crossed polars. Long dimension = 1.4 mm. (f) 67075,48: large feldspar clasts surrounded by fine-grained matrix. Crossed polars. Long dimension = 1.4 mm. (g) 65315,89: medium grain-sized feldspar clasts. Crossed polars. Long dimension = 1.4 mm.

The occurrence of pyroxene as aggregates, heterogeneously distributed, might be explained by the solid-state diffusion into small grains by a large, primary pyroxene grain. The bulk composition of the pyroxene content in these anorthosites suggests pigeonite as the dominant primary pyroxene phase which may have coexisted with a small amount of primary augite. This possibility is also suggested by Hodges and Kushiro (1973).

Sample 60215,13, previously described by Meyer and McCallister (1973), is a polymict breccia, but its grain size and cataclastic texture are similar to Type I textures (see Fig. 1c). The large anorthite clasts display the same shock effects observed in 60025 and are surrounded by a fine-grained matrix. Mineral chemical data suggests that the *bulk* of 60215,13 is a monomict breccia. This suggestion is in accord with Meyer and McCallister (1973). Pyroxene occurs as aggregates and no exsolution textures were observed. Pyroxene is less abundant than in 60025. Lithic clasts were described by Meyer and McCallister as either anorthositic or troctolitic.

Table 2. Modal analyses.

Vol.%						
	Feld	Opx	*	Cpx	Opaques	Opx/Opx + Cpx
60015,113	98.4		1.5		0.1	
,122	98.5		1.3		0.15	
,123	98.8		1.1		0.1	
60025,110	99.0	0.9		0.1	0.05	0.90
,117	98.1	1.7		0.15	0.1	0.92
,120	99.4		0.5		0.1	
,129	99.2	0.6		0.05	0.15	0.95
60215,113	99.6	0.4		0.05	< 0.05	~ 0.9
64819,3	72	1.0		0.1	0.1	0.89
	(brown glass = 27)					
65315,89	98.5	1.2		0.2	0.3	0.85

*Opx and cpx not differentiated.

Apollo 15 anorthosite 15415 has been extensively described by other workers. See, for example, Hargraves and Hollister (1972), James (1972), and Stewart *et al.* (1972). 15415 differs from 60025 and the rest of the samples reported here because of the dominance of augite over opx, however, the textures and relationships in 15415 classify it as Type I.

Type II

Severe shock damage and large-scale local melting characterize Type II textures. Most monomict anorthosites and a large number of the anorthositic coarse fines are classified as Type II. The Type II texture is interpreted as either the reflection of a more complex history after initial excavation than Type I anorthosites or as a more severe initial excavation shock.

60015 sections 113,22,23 display uniform textures (Fig. 1d). Most of the surface of this rock is covered by divitrified glass which was sampled by two of the sections. Sclar *et al.* (1973) described the shock features of the feldspar in 60015,127 and a later paper by Sclar and Bauer (1974) expands this description. Anorthite feldspar comprises over 98% of all three sections of 60015. Grain size is mainly bimodal with large 1–3-mm grains of feldspar set in a fine-grained matrix. Twinning is less frequent than in Type I feldspar. Optical extinction is extremely undulose and patchy, and birefringence is increased in the most severely strained grains. The large grains are anhedral, roughly equant, and rounded. The smallest grains are polygonal and show traces of recrystallization. The large fractures are

usually glass filled and cut all grain sizes. Localized melting is a dominant feature both in the groundmass and in the large grains.

The pyroxene occurs as equilibrated opx and augite with grain size and distribution similar to Type I pyroxenes. Pyroxene grains are anhedral and rounded and show much more shock damage than those of Type I. A few orthopyroxenes contain augite exsolution lamellae and some orthopyroxenes occur as long, thin stringers along feldspar. As in Type I textures, most pyroxene is found in aggregates in the matrix associated with ilmenite, Cr spinel, and troilite. No olivine was found.

The dark-brown, devitrified, glassy coating surrounding 60015 is anorthositic in composition, and its bulk composition as determined by electron microprobe is similar to Apollo 16 soils (see Table 6). Small clasts of feldspar are trapped in the glass, and smaller opaque spheres were probed with the energy non-dispersive detector. Qualitative results indicated large amounts of Fe, Ni, Co, and P in apparent order of decreasing abundance. The boundary between glass and anorthosite is sharp and some of the brown glass has penetrated several large fractures. A splash of melted regolith material seems a reasonable explanation and this hypothesis was also suggested by Sclar and Bauer (1974).

Sample 61016,217 is $\frac{3}{4}$ dark matrix breccia and $\frac{1}{4}$ shocked anorthositic breccia. The type II portion consists of several large feldspar grains of uniform composition which were partially shock melted. These feldspars are now almost isotropic, but twinning is still observable. Several grains of pyroxene were observed in the matrix between these large grains and pyroxene pairs displayed two distinct Fe/Fe + Mg ratios, consistent with the polymict nature of this rock.

Other Type II anorthosites include 64819,3 and several of the 2–4 mm coarse fines (63503,13,12, and 21; and 64503,1,36). 64819,3 (Fig. 1e) is a small area thin section with a single large, severely damaged feldspar grain comprising almost half of the section. Two large, glassy veins which contain brown glass and small grains of feldspar cut across the entire section. The composition of the glass is identical to the feldspar in the rest of the sample.

Type III

Equilibrated, anorthositic breccias comprise this third textural type. Well-documented sample 67075 (Fig. 1f) (Peckett and Brown, 1973; McCullum *et al.*, 1974), 65315 (Fig. 1g), 60215, and 2–4-mm samples 64503,1,26 and 66043,1,4 are characteristic of Type III. Type III samples are generally finer grained than Type I and II and feldspar is well rounded to polygonal. Extensive recrystallization of the fine-grained matrix has resulted in uniform extinction and no observable shock damage for the smallest feldspar grains. The large grains show less shock damage than Types I or II, but fractures and higher birefringence attest to a shock episode.

Pyroxene occurs mainly in the fine-grained matrix but is rarely included within the feldspar. Some opx grains have augite exsolution lamellae. Type II pyroxenes generally have subhedral to almost euhedral grain shapes. Pyroxene compositions are variable within each breccia, but equilibrated opx–augite pairs are still present.

A few grains of olivine were observed petrographically in 67075,48 by Peckett and Brown (1973) and verified in the present study by microprobe analysis. Olivine was also observed in 65315,89.

PYROXENE CHEMISTRY

Although the pyroxenes comprise less than 2 mole% of the anorthosites under consideration they are the most powerful petrogenetic indicators in these rocks. All of the large anorthosites that we examined contain augite–low-calcium pyroxene pairs and in all cases where the low-calcium pyroxenes were large enough for accurate optical determination they were found to be orthopyroxenes. Table 2 presents modal data on some of the anorthosites we considered. An important feature of the modal data is the opx/opx + cpx ratio, which is ≈ 0.9 . With this ratio known and the tie-lines established (Figs. 2 and 3) we can estimate

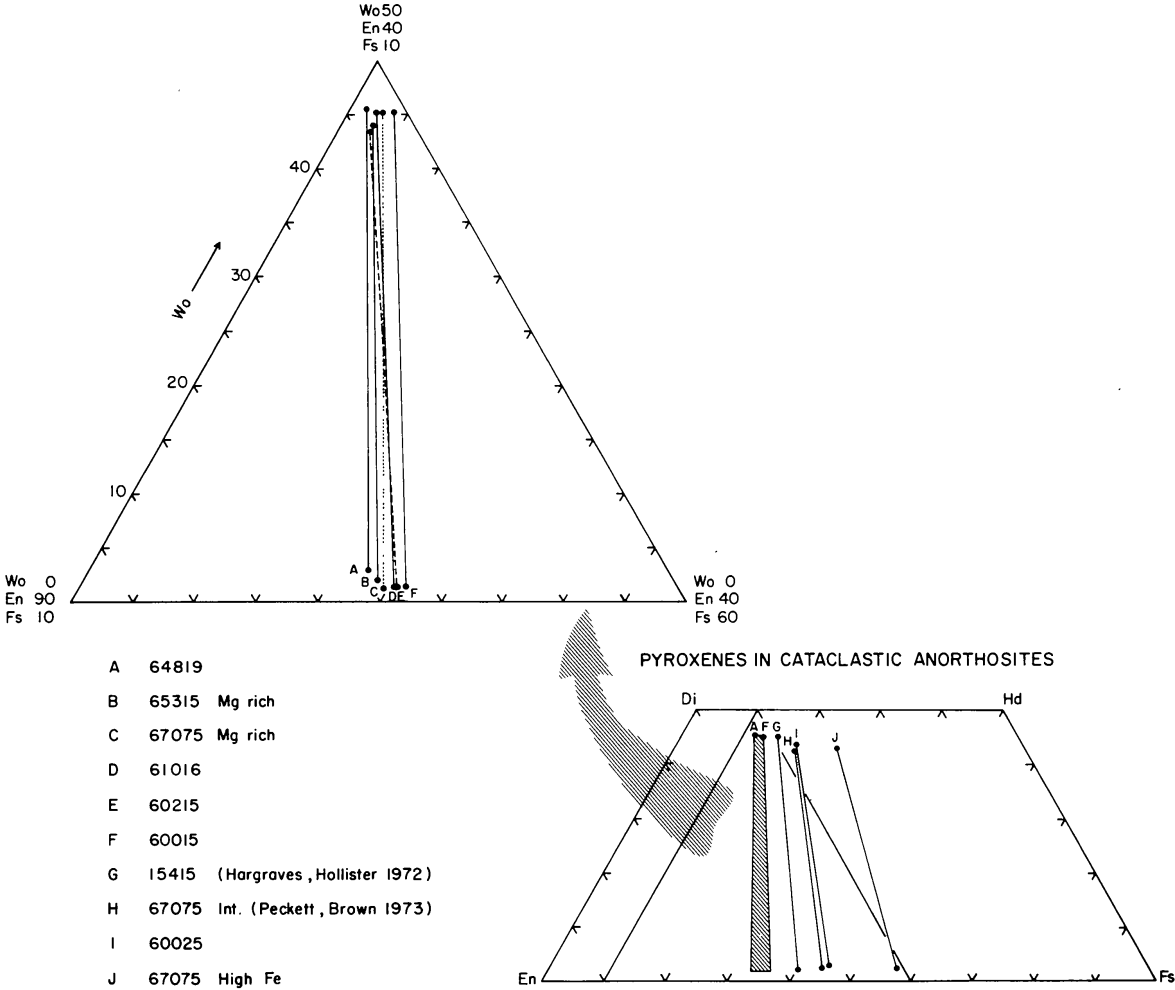


Fig. 2. Pyroxene quadrilateral plots of equilibrated augite–orthopyroxene pairs showing tie-line orientations for the large Apollo 16 anorthosites and anorthositic breccias. “Fs” component includes Mn^{2+} .

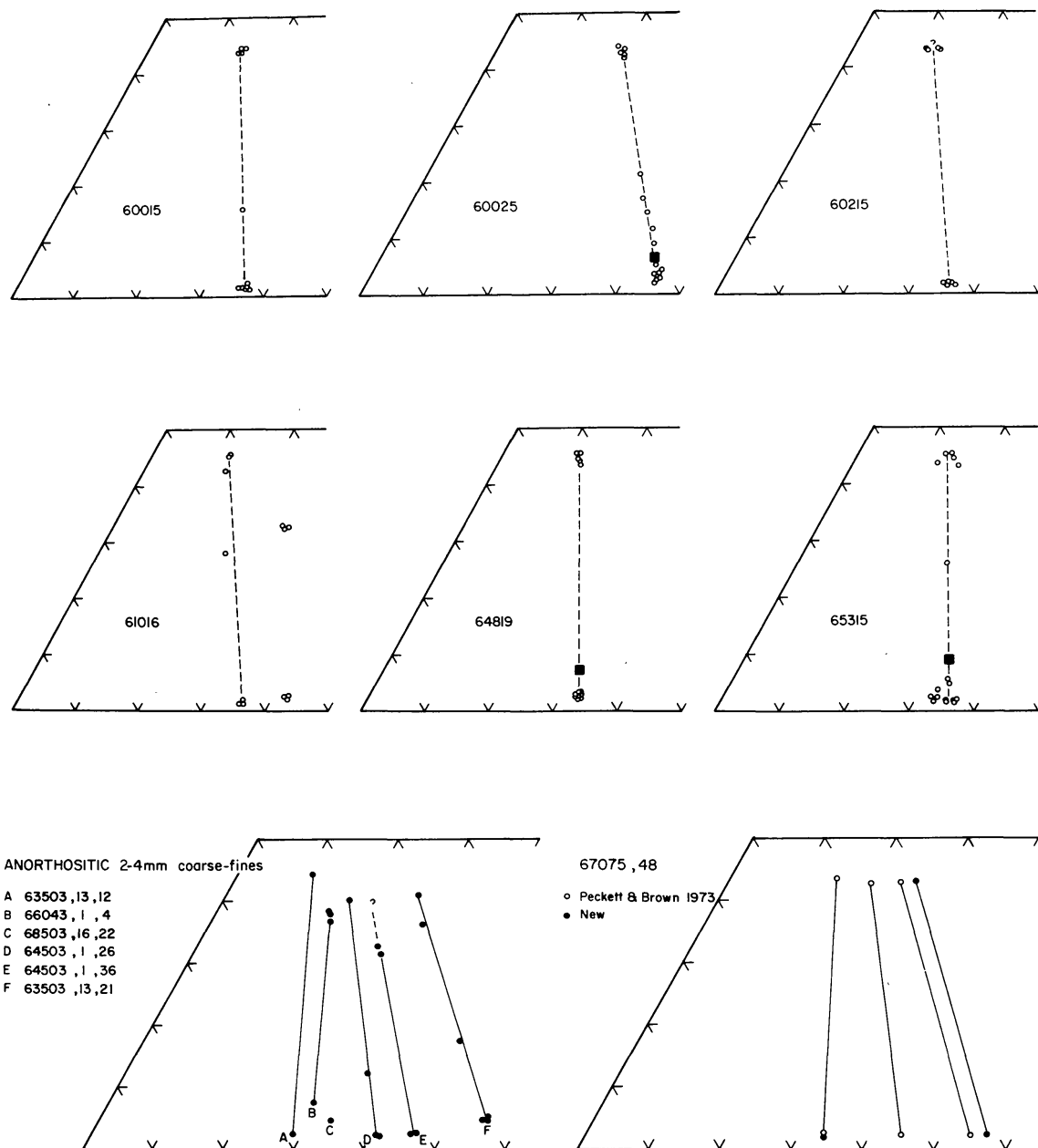


Fig. 3a, b. Pyroxene quadrilaterals for separate rocks including 2-4-mm coarse fines.
 "Fs" component includes Mn^{2+} .

the composition of the primary pyroxene before exsolution, if the present proportions of augite and opx result from the breakdown of a *single* pyroxene phase. The data presented here suggests that the dominant primary pyroxene was a pigeonite that has since inverted to orthopyroxene through a series of exsolution and inversion reactions. In most cases the phase separation is nearly complete and only discrete grains of augite and orthopyroxene remain. These rocks are thus different in one very important aspect from 15415 where the dominant primary pyroxene appears to have been augite. They differ also from those portions of anorthositic breccia 67075 that have high-mafic abundances, well-defined exsolu-

tion lamellae, and which apparently initially crystallized more equal amounts of coexisting augite and pigeonite.

The chemical characteristics of the pyroxenes in the anorthosites are illustrated in Figs. 2–4 and Tables 3 and 4. Figure 2 summarizes our data for coexisting pyroxenes in the Apollo 16 anorthosites. It is clear from this diagram that considerable subsolidus phase separation has occurred between the pyroxenes. Note that most of the pyroxene pairs cluster near the magnesium-rich end of tie-line spectrum, an important point that will be elaborated below in our discussion of petrogenesis. Tables 3 and 4 show that the components other than Wo–En–Fs, “Others,” in the augites range from 3% to 6%, while the “Others” content of the orthopyroxenes ranges from 0.7 to 3%. Since concentration of these “Others” components are relatively low, it appears reasonable to use the data of Lindsley *et al.* (1974) for the 810°C isotherm for coexisting augite and hypersthene. By comparing compositions of coexisting pyroxenes with those of Lindsley *et al.* (1974), we conclude that the Apollo 16 anorthosites considered in this study have pyroxene pairs that equilibrated to temperatures of less than 810°C.

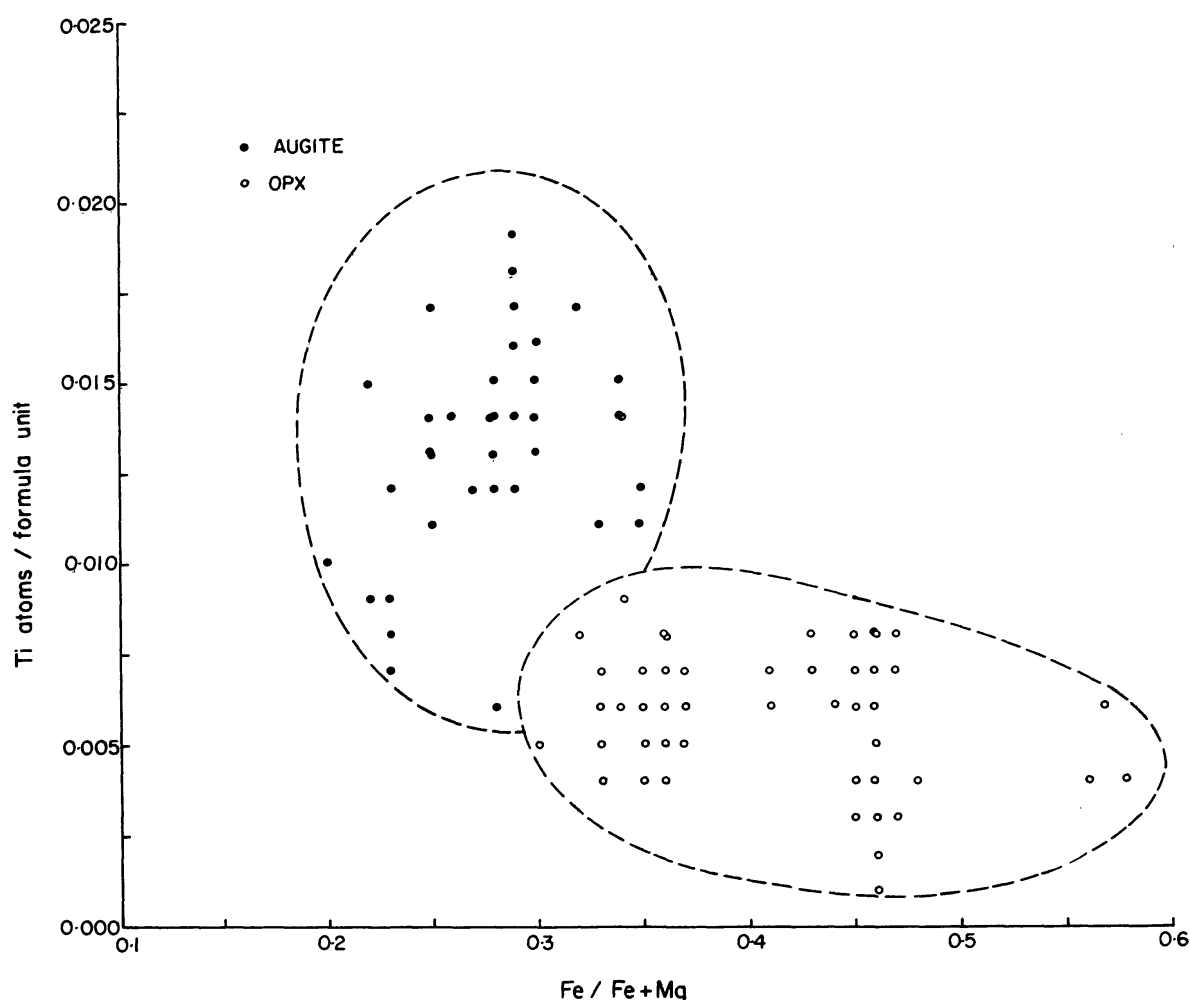


Fig. 4 (a) Ti^{4+} versus $\text{Fe}/\text{Fe} + \text{Mg}$ for opx and augite from all anorthosites.

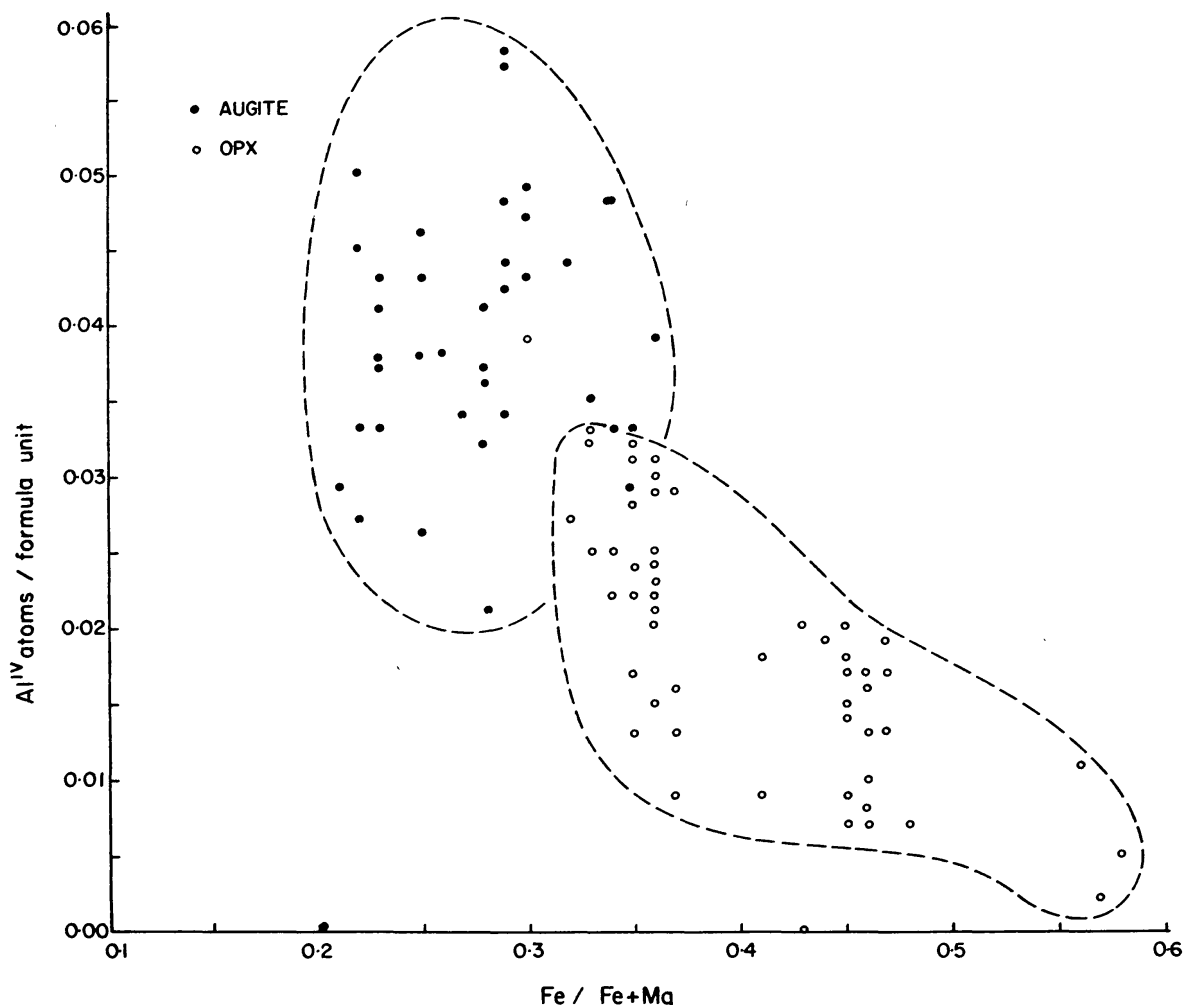


Fig. 4 (b) Tetrahedral Al vs Fe/Fe + Mg for opx and augite from all anorthosites.

Although the “Others” content of pyroxenes in these rocks is low, it is still of interest to define their crystal-chemical character. If we assume that Na, Fe^{3+} , Ti^{3+} , and Cr^{2+} are in such low concentration that they can be ignored in these pyroxenes (which may or may not be a good assumption), we can write a charge balance equation for the non-quadrilateral components as:

$$\text{Charge excess relative to quad} = \text{Charge deficiency relative to quad}$$

$${}^{\text{VI}}\text{Al} + {}^{\text{VI}}\text{Cr}^{3+} + 2{}^{\text{VI}}\text{Ti}^{4+} = {}^{\text{IV}}\text{Al}$$

Thus, if the assumptions stated above are correct, the most important pyroxene “Others” end-members are $\text{R}^{2+}\text{Ti}^{4+}\text{Al}_2\text{O}_6$, $\text{R}^{2+}\text{AlSiAlO}_6$ and $\text{R}^{2+}\text{Cr}^{3+}\text{SiAlO}_6$. In the augites studied, these three components appear present in the following approximate proportions: $\text{CaTi}^{4+}\text{Al}_2\text{O}_6$ (50%), $\text{CaCr}^{3+}\text{SiAlO}_6$ (30%), and $\text{CaAl}^{\text{VI}}\text{SiAlO}_6$ (20%). In orthopyroxene the proportions are $(\text{Mg}, \text{Fe})\text{Ti}^{4+}\text{Al}_2\text{O}_6$ (48%), $(\text{Mg}, \text{Fe})\text{Cr}^{3+}\text{SiAlO}_6$ (26%), and $(\text{Mg}, \text{Fe})\text{AlSiAlO}_6$ (26%). Thus, the most abundant substitutional couple in both augite and opx is ${}^{\text{VI}}\text{Ti}^{\text{IV}}\text{Al}_2$ with lesser amounts of ${}^{\text{VI}}\text{Cr}^{3+\text{IV}}\text{Al}$

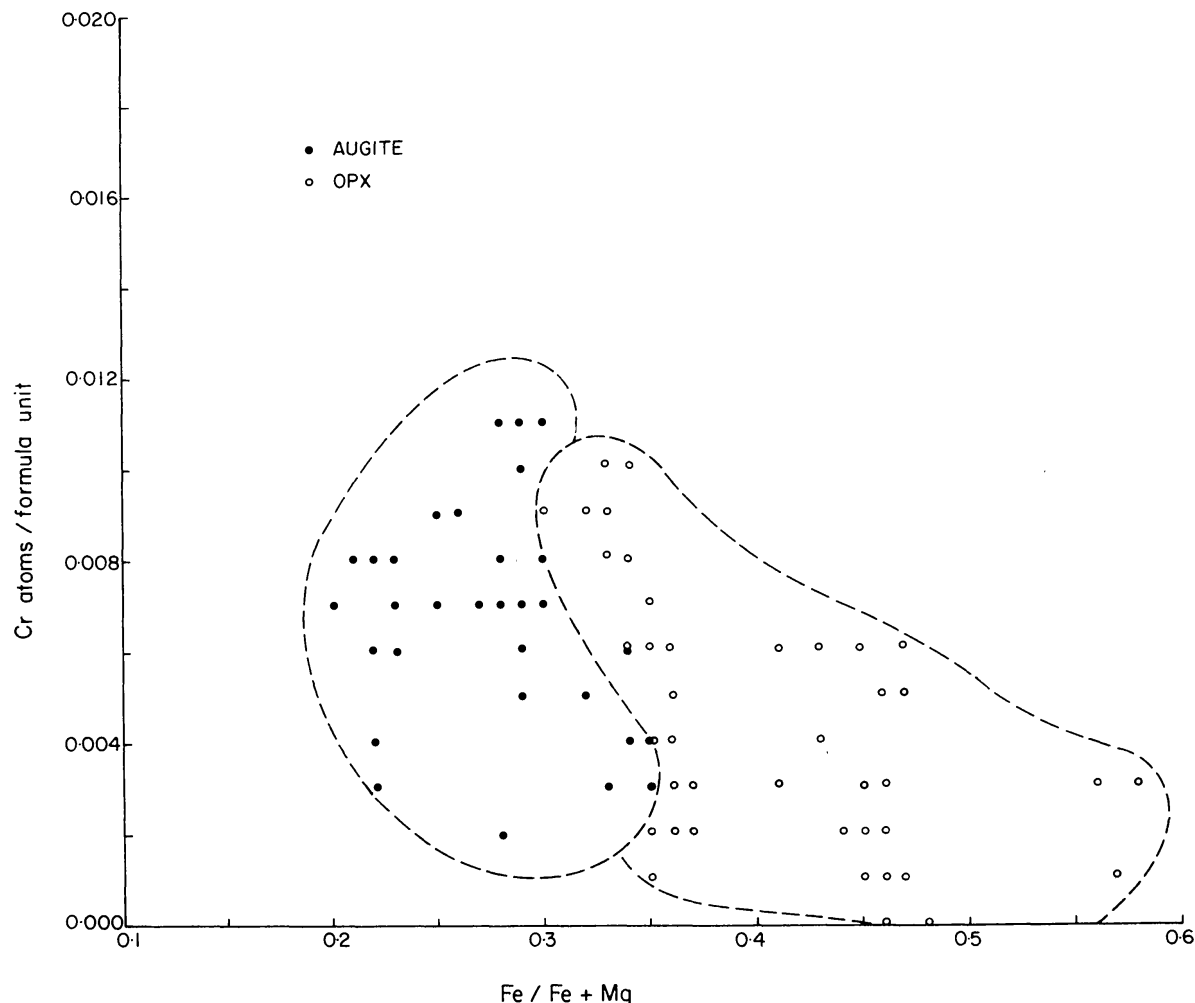


Fig. 4 (c) Cr^{3+} versus $\text{Fe}/\text{Fe} + \text{Mg}$ for opx and augite from all anorthosites.

and $^{\text{VI}}\text{Al}^{\text{IV}}\text{Al}$. Although the character of the “Others” components in both augite and opx are similar, their abundances are not. Figures 4a, 4b, and 4c show that augite is enriched in Ti, Al, and Cr relative to orthopyroxene. Thus, for these highly equilibrated pyroxenes ($T < 800^\circ\text{C}$) the components $\text{CaTiAl}_2\text{O}_6$, CaAl-SiAlO_6 , and $\text{CaCr}^{3+}\text{SiAlO}_6$ appear to have greater stability than their orthorhombic (Mg, Fe) equivalents. Figure 4d shows that Mn is enriched in opx relative to augite as would be expected since the Mn would probably be largely ordered into the pyroxene M2 site which is mainly occupied by Ca in augite.

Figure 3 summarizes the pyroxene tie-line orientations for our Apollo 16 large anorthosites and 2–4-mm coarse fine fragments. The observed tie-line sweep for our samples is similar to that observed for polymict breccia 67075 and appears to be fairly representative of the anorthosites at the Apollo 16 site. We believe that chemical and modal data for these rocks suggests that they were all derived from the same layered complex and that pyroxene tie-lines indicate progressive crystallization (left to right) on the pyroxene quadrilaterals.

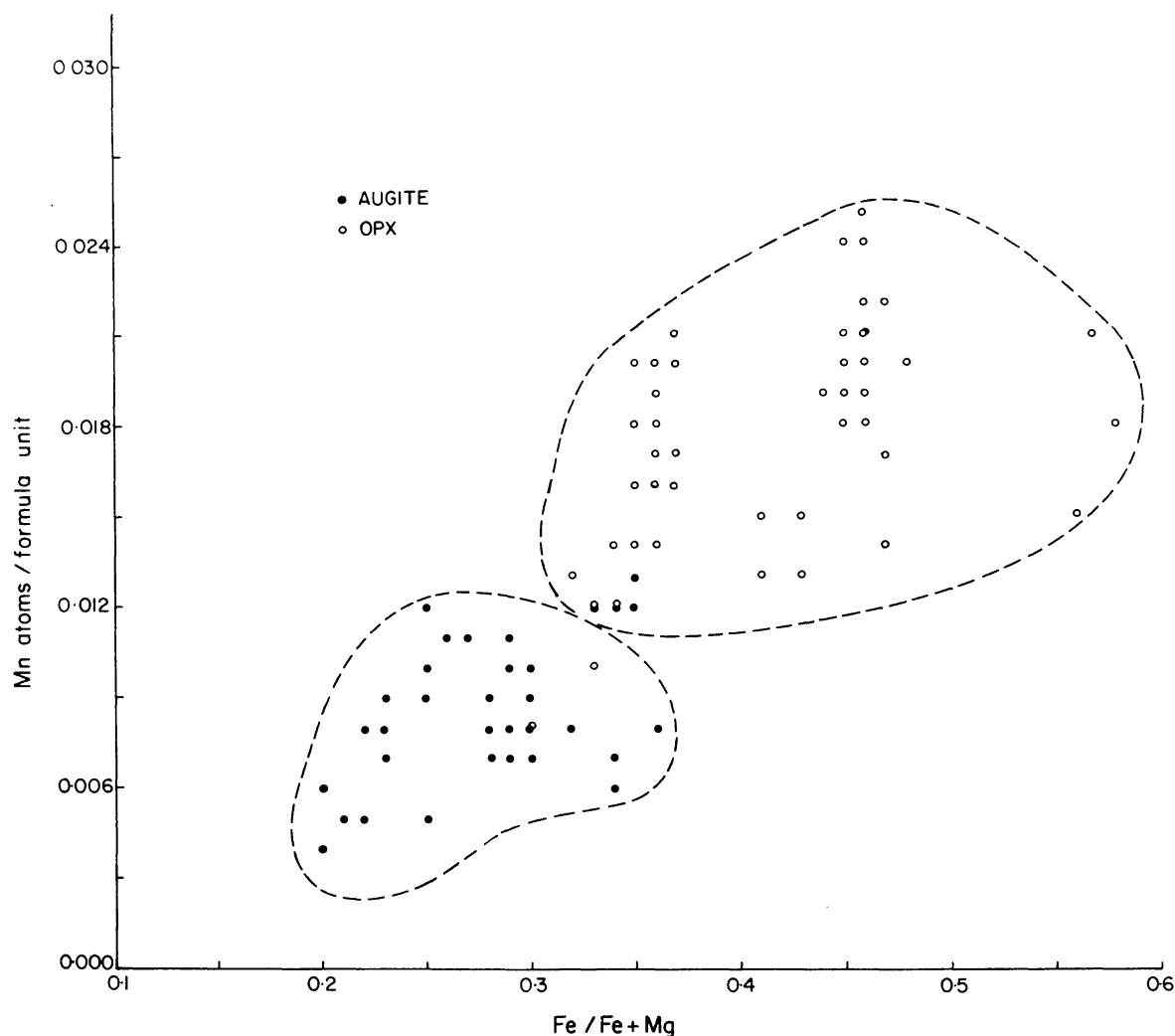


Fig. 4 (d) Mn^{2+} versus $\text{Fe}/\text{Fe} + \text{Mg}$ for opx and augite from all anorthosites. (Note: all quadrilateral plots include Mn^{2+} with Fe^{2+} .)

FELDSPAR-PYROXENE RELATIONS

Table 5 summarizes our chemical data for plagioclase feldspars from the Apollo 16 anorthosites. The An content ranges from 97.5% to 96%, MgO has the range 0–0.2 wt.% and FeO has the range 0.05–0.30 wt.%. Although the chemical variations in the feldspar are very small, there does appear to be some interesting systematics for the Ca, Mg, and Fe partitioning between feldspar and pyroxene. Figure 5 suggests that there is some correlation of An% in feldspar with the $\text{Fe}/\text{Fe} + \text{Mg}$ ratio of opx. Although there are obviously large errors connected with these small variations the trends does appear to be the normal one expected for igneous differentiation.

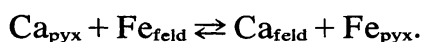
Figure 6 shows increasing wt.% MgO + FeO in the feldspar with increasing $\text{Fe}/\text{Fe} + \text{Mg}$ of the opx in the Apollo 16 anorthosites. This trend is apparently a

Table 3. Orthopyroxene microprobe

	60015			60025		
	1	2	3	4	5	6
SiO ₂	53.40	53.00	51.80	51.80	51.40	51.40
Al ₂ O ₃	0.49	0.51	0.66	0.35	0.16	0.19
TiO ₂	0.24	0.24	0.20	0.20	0.14	0.28
Cr ₂ O ₃	0.07	0.09	0.06	0.08	0.00	0.06
FeO	22.50	22.60	22.80	27.70	27.60	28.10
MnO	0.52	0.62	0.53	0.66	0.68	0.56
MgO	21.90	21.90	21.80	18.60	18.20	18.90
CaO	0.88	0.80	1.21	1.30	1.48	1.04
Na ₂ O	0.00	0.00	0.00	0.00	0.00	0.00
Σ	100.00	99.76	99.06	100.69	99.66	100.53
Si	1.991	1.984	1.962	1.974	1.981	1.966
^{iv} Al	0.009	0.016	0.029	0.016	0.007	0.009
Σ Tet	2.000	2.000	1.991	1.990	1.988	1.975
Ti	0.007	0.007	0.006	0.006	0.004	0.008
^{vi} Al	0.012	0.006	0.000	0.000	0.000	0.000
Cr	0.002	0.003	0.002	0.002	0.000	0.002
Fe	0.701	0.707	0.722	0.883	0.890	0.899
Mn	0.016	0.020	0.017	0.021	0.022	0.018
Mg	1.217	1.222	1.230	1.056	1.045	1.077
Σ Oct	1.955	1.965	1.977	1.968	1.961	2.004
Ca	0.035	0.032	0.049	0.053	0.061	0.043
Na	0.000	0.000	0.000	0.000	0.000	0.000
Σ Total Cations	3.990	3.997	4.017	4.011	4.010	4.022
Quad	97.9	98.4	97.1	98.4	99.3	99.0
Others	2.1	1.6	2.9	1.6	0.7	1.0
Breakdown of Quad						
Ca	1.8	1.6	2.5	2.7	3.1	2.1
Mg	62.3	62.3	61.5	53.0	52.4	53.4
Fe	35.9	36.1	36.1	44.3	44.6	44.5
Fe/Fe + Mg	0.37	0.37	0.37	0.46	0.46	0.45

result of progressive differentiation and adds further support to the cogenetic nature of the Apollo 16 anorthosites.

Metamorphic equilibration between pyroxene and feldspar is indicated by the data presented in Fig. 7. Microprobe traverses through feldspar away from pyroxene were performed on the most pristine sample available, 60025. Large relatively undamaged feldspar grains adjacent to an orthopyroxene and to an augite were analyzed in 5-μ steps. Each step was analyzed 5–8 times and only average values are plotted in Fig. 7. Obvious differences are indicated between feldspar equilibrium with opx and augite. A simple exchange reaction seems apparent:



1975LPSC.....6...263D

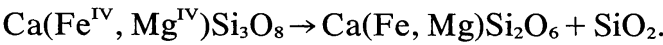
analyses from large Apollo 16 anorthosites.

60215		61016			64819		65315		
7	8	9	10	11	12	13	14	15	16
52.60	53.35	53.60	52.60	52.10	52.20	52.97	52.80	52.90	53.20
0.54	0.80	0.45	0.45	0.39	0.57	0.74	0.50	0.72	0.67
0.08	0.21	0.18	0.18	0.18	0.14	0.21	0.23	0.26	0.28
0.12	0.20	0.19	0.19	0.15	0.29	0.36	0.19	0.23	0.32
22.00	22.30	22.40	22.40	21.90	20.60	21.17	21.20	22.20	20.00
0.63	0.58	0.53	0.53	0.49	0.32	0.32	0.43	0.45	0.40
22.80	22.80	22.40	22.40	22.60	23.80	24.03	23.60	23.40	23.50
0.74	0.79	0.87	0.87	0.81	1.51	1.71	1.18	0.85	1.21
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
99.51	101.03	100.62	99.62	98.62	99.43	101.51	100.13	101.01	99.58
1.971	1.968	1.985	1.973	1.971	1.951	1.943	1.961	1.954	1.973
0.024	0.032	0.015	0.020	0.017	0.025	0.032	0.022	0.031	0.027
1.995	2.000	2.000	1.993	1.988	1.976	1.975	1.983	1.985	2.000
0.002	0.006	0.005	0.005	0.005	0.004	0.006	0.006	0.007	0.008
0.000	0.003	0.005	0.000	0.000	0.000	0.000	0.000	0.000	0.002
0.004	0.006	0.006	0.006	0.004	0.009	0.010	0.006	0.007	0.009
0.689	0.688	0.694	0.703	0.693	0.644	0.649	0.658	0.686	0.620
0.020	0.018	0.017	0.017	0.016	0.010	0.010	0.014	0.014	0.013
1.273	1.254	1.236	1.252	1.274	1.326	1.313	1.306	1.288	1.299
1.988	1.974	1.963	1.982	1.992	1.993	1.989	1.990	2.002	1.951
0.030	0.031	0.035	0.035	0.033	0.060	0.067	0.047	0.034	0.048
0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
4.013	4.005	3.998	4.010	4.013	4.029	4.030	4.020	4.021	3.999
97.6	96.8	98.4	98.0	98.3	97.5	96.8	97.8	96.9	97.3
2.4	3.2	1.6	2.0	1.7	2.5	3.2	2.2	3.1	2.7
1.5	1.6	1.8	1.8	1.6	3.0	3.3	2.3	1.7	2.4
63.9	63.5	62.9	62.9	63.7	65.3	64.7	64.9	64.2	66.0
34.6	34.9	35.3	35.3	34.6	31.7	32.0	32.7	34.2	31.5
0.35	0.35	0.36	0.36	0.35	0.33	0.33	0.34	0.35	0.32

This would explain the approximate mirror imaging of the Ca and Fe trends in both figures. The different partitioning between feldspar and opx and between feldspar and augite would be expected because of the differences in crystal structure of these two pyroxenes which especially affects the amount of calcium the pyroxenes can accommodate. It should also be noted that this is a mechanism whereby the pyroxenes become iron enriched during metamorphism. Smith and Steele (1974) suggest a different mechanism which is discussed below.

Similar results were arrived at during reconnaissance traverses of the same relationships in sample 60015. The results are not sufficiently quantitative for presentation, but it is obvious that some sort of cation diffusion took place between pyroxene and feldspar.

Smith and Steele (1974) have proposed an interesting hypothesis to account for the iron-rich nature of many of the pyroxenes in anorthosites. They hypothesize that some of these pyroxenes are exsolution products from solid solutions involving pyroxene components in feldspars. The type of reaction they propose is:



This apparently indicates that Fe and Mg would be accommodated in the feldspar tetrahedral sites. They point out that if this process took place, the ultimate Fe/Mg ratio would be determined by crystal-liquid equilibria involving liquid and plagioclase. The distribution coefficients are such that plagioclase is considerably more enriched in Fe relative to a silicate melt than is pyroxene (Smith and Steele,

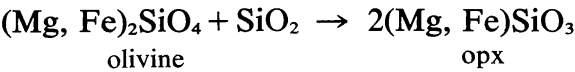
Table 4. Clinopyroxene microprobe analyses from large Apollo 16 anorthosites.

	60015			60025			60215		
	1	2	3	4	5	6	7	8	9
SiO ₂	52.40	51.80	52.90	52.50	51.30	51.80	54.50	53.01	53.20
Al ₂ O ₃	0.87	1.13	1.07	0.65	1.08	0.73	1.07	0.95	0.98
TiO ₂	0.48	0.47	0.50	0.38	0.51	0.42	0.38	0.30	0.32
Cr ₂ O ₃	0.24	0.30	0.30	0.14	0.20	0.11	0.23	0.25	0.23
FeO	8.90	8.80	9.10	12.60	11.90	12.40	6.80	8.36	8.20
MnO	0.29	0.32	0.34	0.38	0.21	0.39	0.20	0.28	0.27
MgO	14.90	14.50	14.70	13.30	13.10	13.20	15.70	15.71	15.70
CaO	21.50	21.60	21.60	20.80	21.00	20.80	21.00	21.82	21.80
Na ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Σ	99.58	98.92	100.51	100.45	99.30	99.85	99.88	100.68	100.70
Si	1.962	1.954	1.962	1.967	1.952	1.963	2.000	1.958	1.962
^{IV} Al	0.038	0.046	0.038	0.029	0.048	0.033	0.000	0.041	0.038
Σ Tet	2.000	2.000	2.000	1.996	2.000	1.996	2.000	1.999	2.000
Ti	0.014	0.013	0.014	0.011	0.015	0.012	0.010	0.008	0.009
^{VI} Al	0.000	0.004	0.009	0.000	0.001	0.000	0.046	0.000	0.004
Cr	0.007	0.009	0.009	0.004	0.006	0.003	0.007	0.007	0.007
Fe	0.279	0.278	0.282	0.397	0.379	0.393	0.209	0.258	0.253
Mn	0.009	0.010	0.011	0.012	0.007	0.013	0.006	0.009	0.008
Mg	0.831	0.815	0.813	0.747	0.743	0.746	0.859	0.865	0.863
Σ Oct	1.140	1.129	1.138	1.171	1.151	1.167	1.137	1.147	1.144
Ca	0.862	0.873	0.859	0.840	0.856	0.845	0.826	0.863	0.861
Na	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Σ Total Cations	4.002	4.003	3.996	4.006	4.006	4.007	3.963	4.010	4.005
Quad	96.2	95.4	96.2	97.1	95.2	96.7	93.7	95.9	96.2
Others	3.8	4.6	3.8	2.9	4.8	3.3	6.3	4.1	3.8
Breakdown of Quad									
Ca	43.7	44.4	43.9	42.3	43.3	42.6	43.6	43.5	43.6
Mg	42.1	41.5	41.6	37.6	37.6	37.6	45.4	43.5	43.6
Fe	14.1	14.1	14.5	20.0	19.1	19.8	11.0	13.0	12.8
Fe/Fe + Mg	0.25	0.25	0.26	0.35	0.34	0.35	0.20	0.23	0.23

Table 4. (Continued).

	61016			64819			65315		
	10	11	12	13	14	15	16	17	18
SiO ₂	53.20	53.00	51.38	53.00	52.80	52.10	52.70	52.90	52.24
Al ₂ O ₃	1.01	1.00	1.32	0.85	1.04	1.53	1.03	0.99	1.04
TiO ₂	0.45	0.43	0.42	0.23	0.09	0.55	0.26	0.49	0.55
Cr ₂ O ₃	0.25	0.27	0.25	0.27	0.22	0.26	0.26	0.37	0.38
FeO	10.10	10.00	10.40	7.50	7.80	7.60	8.10	9.90	10.96
MnO	0.27	0.26	0.23	0.16	0.17	0.17	0.24	0.30	0.31
MgO	14.30	14.40	14.60	15.50	15.80	15.20	14.90	14.50	14.57
CaO	21.50	21.60	21.50	22.10	22.40	21.50	22.40	21.30	19.98
Na ₂ O	0.00	0.00	0.00	0.00	0.00	0.04	0.00	0.00	0.00
Σ	101.08	100.96	100.10	99.61	100.32	98.95	99.89	100.75	100.03
Si	1.968	1.964	1.930	1.971	1.955	1.950	1.963	1.963	1.957
^{iv} Al	0.032	0.036	0.058	0.029	0.045	0.050	0.037	0.037	0.043
Σ Tet	2.000	2.000	1.988	2.000	2.000	2.000	2.000	2.000	2.000
Ti	0.013	0.012	0.012	0.006	0.003	0.015	0.007	0.014	0.015
^{vi} Al	0.012	0.008	0.000	0.008	0.000	0.018	0.008	0.006	0.003
Cr	0.007	0.008	0.007	0.008	0.006	0.008	0.008	0.011	0.011
Fe	0.312	0.310	0.327	0.233	0.242	0.238	0.252	0.307	0.343
Mn	0.008	0.008	0.007	0.005	0.005	0.005	0.008	0.009	0.010
Mg	0.788	0.795	0.817	0.859	0.872	0.848	0.827	0.802	0.814
Σ Oct	1.140	1.141	1.170	1.119	1.128	1.132	1.110	1.149	1.196
Ca	0.852	0.858	0.865	0.881	0.889	0.862	0.894	0.847	0.802
Na	0.000	0.000	0.000	0.000	0.000	0.003	0.000	0.000	0.000
Σ Total Cations	3.994	3.998	4.023	4.000	4.017	3.998	4.004	3.996	3.998
Quad	96.8	96.4	94.2	97.1	95.5	95.0	96.3	96.3	95.7
Others	3.2	3.6	5.8	2.9	4.5	5.0	3.7	3.7	4.3
Breakdown of Quad									
Ca	43.6	43.7	43.1	44.6	44.4	44.3	45.3	43.3	40.9
Mg	40.4	40.5	40.7	43.5	43.5	43.5	41.9	41.0	41.5
Fe	16.0	15.8	16.3	11.8	12.1	12.2	12.8	15.7	17.5
Fe/Fe + Mg	0.28	0.28	0.29	0.21	0.22	0.22	0.23	0.28	0.30

1974). This is a very interesting hypothesis that requires further testing. We have looked specifically for evidence for and against this idea. One obvious test is the modal proportions of pyroxene and silica. Our studies show that although there is some silica present in these anorthosites, the amounts are so minor relative to pyroxene ($\text{SiO}_2/\text{SiO}_2 + \text{pyroxene} \leq 0.01$) that the exsolution reaction as written above is cast in serious doubt. Of course, it is possible that more silica was present and it has subsequently reacted away, for example, by the reaction:



However, in rocks like 60025 and 60015 that were studied extensively it appears

Table 5. Feldspar microprobe analyses from large Apollo 16 anorthosites.

	60015					60025					60215				
	1	2	3	4	5	6	7	8	9	10	11	12	13		
SiO ₂	44.6	44.8	44.5	44.6	44.6	43.4	44.7	44.8	44.4	44.8	44.5	45.5	44.7		
Al ₂ O ₃	35.2	36.3	36.3	35.9	36.0	35.2	35.2	35.2	35.4	34.4	35.7	35.3	35.9		
FeO	0.20	0.09	0.15	0.10	0.18	0.19	0.15	0.19	0.17	0.23	0.14	0.16	0.23		
MgO	0.06	0.01	0.07	0.05	0.07	0.04	0.02	0.06	0.07	0.02	0.03	0.01	0.02		
CaO	20.0	19.7	19.7	19.4	19.5	19.5	19.6	19.3	19.5	19.6	20.2	19.0	18.9		
Na ₂ O	0.35	0.37	0.32	0.38	0.38	0.42	0.39	0.48	0.46	0.36	0.37	0.44	0.39		
K ₂ O	0.01	0.01	0.00	0.00	0.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		
Σ	100.4	101.3	101.0	100.4	100.8	98.8	100.1	100.0	100.0	99.4	100.9	100.4	99.7		
Si	2.057	2.044	2.035	2.059	2.046	2.034	2.063	2.069	2.053	2.085	2.043	2.088	2.064		
Al	1.915	1.952	1.959	1.931	1.947	1.947	1.918	1.918	1.930	1.885	1.932	1.908	1.956		
Fe	0.008	0.004	0.006	0.004	0.007	0.007	0.006	0.007	0.007	0.009	0.005	0.006	0.009		
Mg	0.004	0.001	0.005	0.003	0.005	0.003	0.001	0.004	0.005	0.001	0.002	0.001	0.002		
Σ	3.984	4.001	4.005	3.997	4.005	3.991	3.988	3.998	3.995	3.980	3.982	4.003	4.031		
Ca	0.987	0.964	0.966	0.961	0.959	0.981	0.972	0.952	0.967	0.977	0.992	0.935	0.936		
Na	0.031	0.033	0.028	0.034	0.034	0.038	0.035	0.043	0.041	0.033	0.033	0.039	0.035		
K	0.001	0.001	0.000	0.000	0.002	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000		
Σ	1.019	0.998	0.994	0.995	0.995	1.019	1.007	0.995	1.008	1.010	1.025	0.974	0.971		
An	96.9	96.6	97.1	96.6	96.4	96.3	96.5	95.7	95.9	96.8	96.8	96.0	96.4		
Ab	3.0	3.3	2.9	3.5	3.4	3.7	3.5	4.3	4.0	3.2	3.2	4.0	3.6		
Or	0.05	0.1	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Table 5. (Continued).

	61016					64819					65315					67075				
	14	15	16	17	18	19	20	21	22	23	24	25	26							
SiO ₂	43.9	43.4	45.3	44.7	44.8	43.2	43.2	42.9	43.2	45.4	42.8	43.3	43.6							
Al ₂ O	35.7	35.5	35.9	36.7	35.4	35.0	34.9	36.3	36.6	34.9	35.6	36.2	35.6							
FeO	0.16	0.15	0.27	0.11	0.15	0.13	0.10	0.17	0.08	0.11	0.10	0.06	0.09							
MgO	0.04	0.11	0.18	0.04	0.06	0.05	0.06	0.01	0.00	0.02	0.00	0.00	0.00							
CaO	19.2	20.0	19.8	19.3	18.9	19.4	18.8	19.6	19.6	19.9	18.9	19.1	19.9							
Na ₂ O	0.45	0.41	0.43	0.35	0.39	0.37	0.36	0.31	0.25	0.33	0.33	0.34	0.31							
K ₂ O	0.01	0.01	0.01	0.06	0.08	0.00	0.04	0.00	0.00	0.00	0.00	0.01	0.01							
Σ	99.5	99.6	101.9	101.3	99.8	98.2	97.5	99.3	99.7	100.7	97.7	99.0	99.5							
Si	2.040	2.024	2.058	2.037	2.080	2.040	2.047	2.002	2.002	2.085	2.024	2.020	2.030							
Al	1.956	1.949	1.918	1.971	1.938	1.944	1.950	1.996	1.999	1.892	1.984	1.991	1.954							
Fe	0.006	0.006	0.010	0.004	0.006	0.005	0.004	0.007	0.003	0.004	0.004	0.003	0.003							
Mg	0.003	0.007	0.012	0.003	0.004	0.003	0.004	0.001	0.000	0.002	0.000	0.000	0.000							
Σ	4.005	3.986	3.998	4.015	4.028	3.992	4.005	4.006	4.004	3.983	4.012	4.014	3.987							
Ca	0.955	0.999	0.965	0.945	0.941	0.979	0.956	0.981	0.973	0.980	0.957	0.955	0.991							
Na	0.041	0.037	0.037	0.031	0.035	0.033	0.033	0.028	0.023	0.029	0.030	0.030	0.028							
K	0.000	0.001	0.001	0.003	0.005	0.000	0.003	0.000	0.000	0.000	0.000	0.001	0.001							
Σ	0.996	1.037	1.003	0.989	0.981	1.012	1.092	1.009	0.996	1.009	0.987	0.986	1.020							
An	95.9	96.4	96.2	96.5	95.8	96.7	96.4	97.2	97.7	97.1	97.0	96.9	97.2							
Ab	4.1	3.5	3.7	3.2	3.7	3.3	3.4	2.8	2.3	2.9	3.0	3.1	2.8							
Or	0.0	0.05	0.1	0.3	0.5	0.0	0.2	0.0	0.0	0.0	0.0	0.05	0.05							

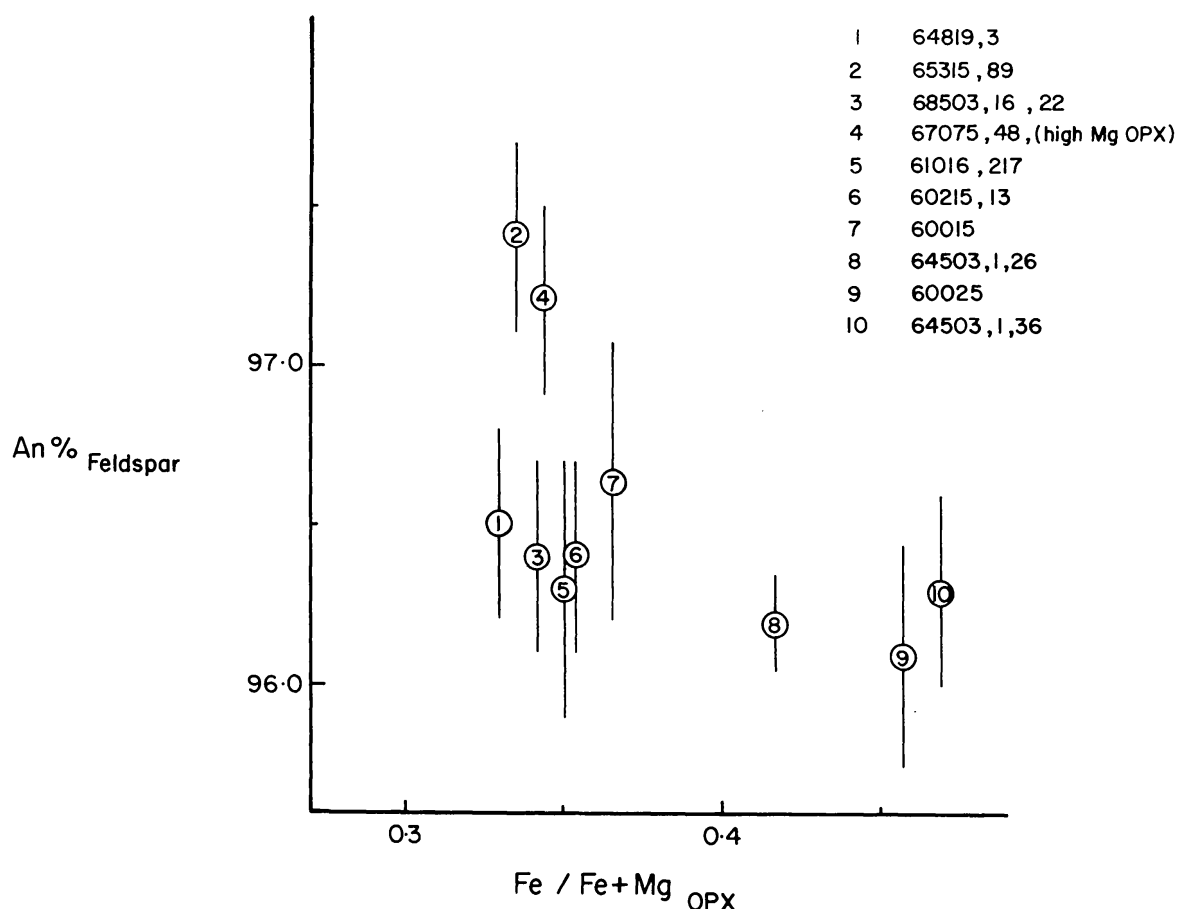


Fig. 5. Anorthite mole% of feldspar versus Fe/Fe+Mg of coexisting opx for ten anorthosites. Anorthite % values are averages for each sample. Error bars equal one standard deviation.

highly fortuitous that all of the hypothetical olivine has been so efficiently removed. This would appear to require that the “exsolved” silica knew just how much olivine would have to be reacted away so that the record of both the olivine and silica could be so efficiently erased. Another problem with the Smith–Steele hypothesis is that they cannot use one mechanism (in the *monomict* anorthosites) for pyroxene that have textures that suggest exsolution and a different mechanism for the interstitial pyroxenes because they have identical compositions. We believe that the feldspar and pyroxene were indeed in communication (Fig. 7) and that exchange of Ca, Fe, and Mg has taken place. However, we conclude that most of this exchange took place during prolonged metamorphism *after* the feldspar and pyroxene *coprecipitated* as *primary phases*. The cation diffusion is such that Fe and Mg are removed from the feldspar and are replaced by Ca that was initially accommodated in the pyroxene structure.

PETROLOGIC HISTORY

A generally accepted model for the origin of lunar anorthosites is a plagioclase cumulate which collected under the crust during the primary differentiation of the

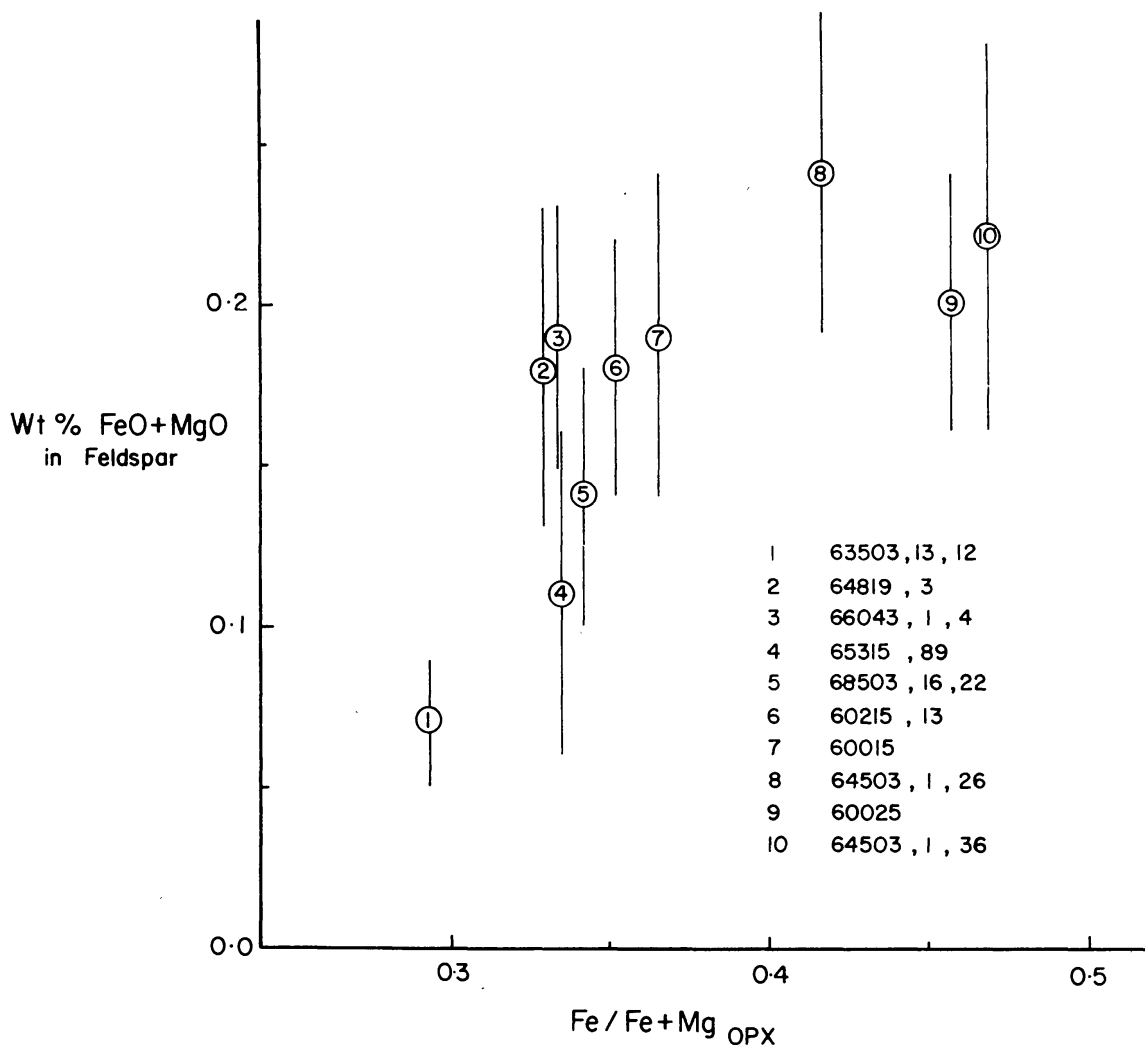


Fig. 6. Wt.% FeO + MgO of feldspar versus Fe/Fe + Mg of coexisting opx for ten anorthosites. Wt.% MgO + FeO values are averages. Error bars equal one standard deviation.

moon. See, for example, Smith *et al.* (1970), Wood *et al.* (1970), and Prinz *et al.* (1973). The Taylor-Jakeš (1974) model for lunar differentiation offers a reasonable explanation for the origin and chemistry of lunar anorthosites and hypothesizes a plagioclase cumulate between the depths of 10 and 40 km. The feldspar cumulate would have a low-albite component and any trapped liquid or crystals would be expected to display Fe/Fe + Mg ratios higher than the average highlands ratio (~ 0.3) since the highlands represent a more primitive magma compared to the fractionated anorthosite. The Apollo 16 anorthosites are consistent with these predictions: feldspar is An_{95-100} and pyroxene Fe/Fe + Mg ratios are generally greater than 0.3.

The continuous range of Fe/Fe + Mg ratios for the trapped pyroxenes and the range in FeO + MgO of the feldspar is appropriate for a large-scale layered, cumulate complex (see Fig. 5). The lithology of this simple model would put the most Fe and Na rich anorthosites at greater depths than those lowest in albite or ferrosilite components. The primary cumulate textures and mineralogy of the

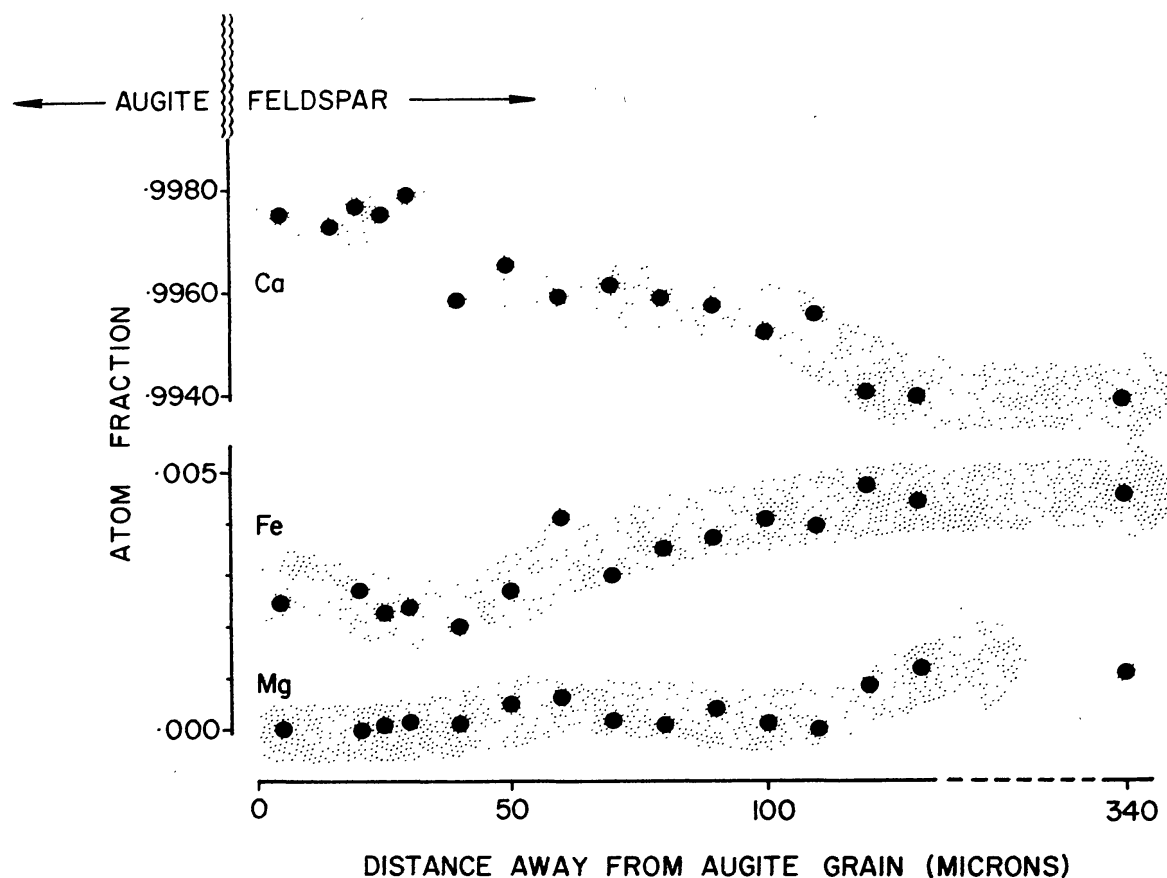


Fig. 7 (a) Diffusion profile: average Ca, Fe, and Mg concentrations in feldspar versus distance from coexisting augite. Shaded band is approximately two standard deviations wide.

Apollo 16 anorthosites has been obscured by metamorphism, but primary mineralogy was probably anorthite, pigeonite with perhaps minor augite, ilmenite, and Cr spinel. Of course, any primary pigeonite would have inverted to opx with augite exsolution during the final cooling stages. Even the relict inverted pigeonite textures would be erased by prolonged metamorphism. The bulk pyroxene compositions are consistent with some primary pigeonite (see Fig. 3a) and examples of augite lamellae in opx hosts are observable in many of the anorthosites.

After accumulating at 10–40 km depths, the anorthosites probably began an extremely slow cooling process. It was during this period that the metamorphic character of the anorthosites was probably developed. Any traces of Ab–An zoning in the feldspar grains have been lost due to solid-state diffusion to uniform compositions. Pigeonite exsolved augite and then inverted to orthopyroxene during this slow cooling. Migration of augite lamellae in the solid-state created larger augite grains as well as exptaxial concentrations on opx grains (see Fig. 1d). The observed pyroxene equilibration, when compared to the synthetic pyroxene isotherm by Lindsley *et al.* (1974), leads to temperatures of equilibration of less

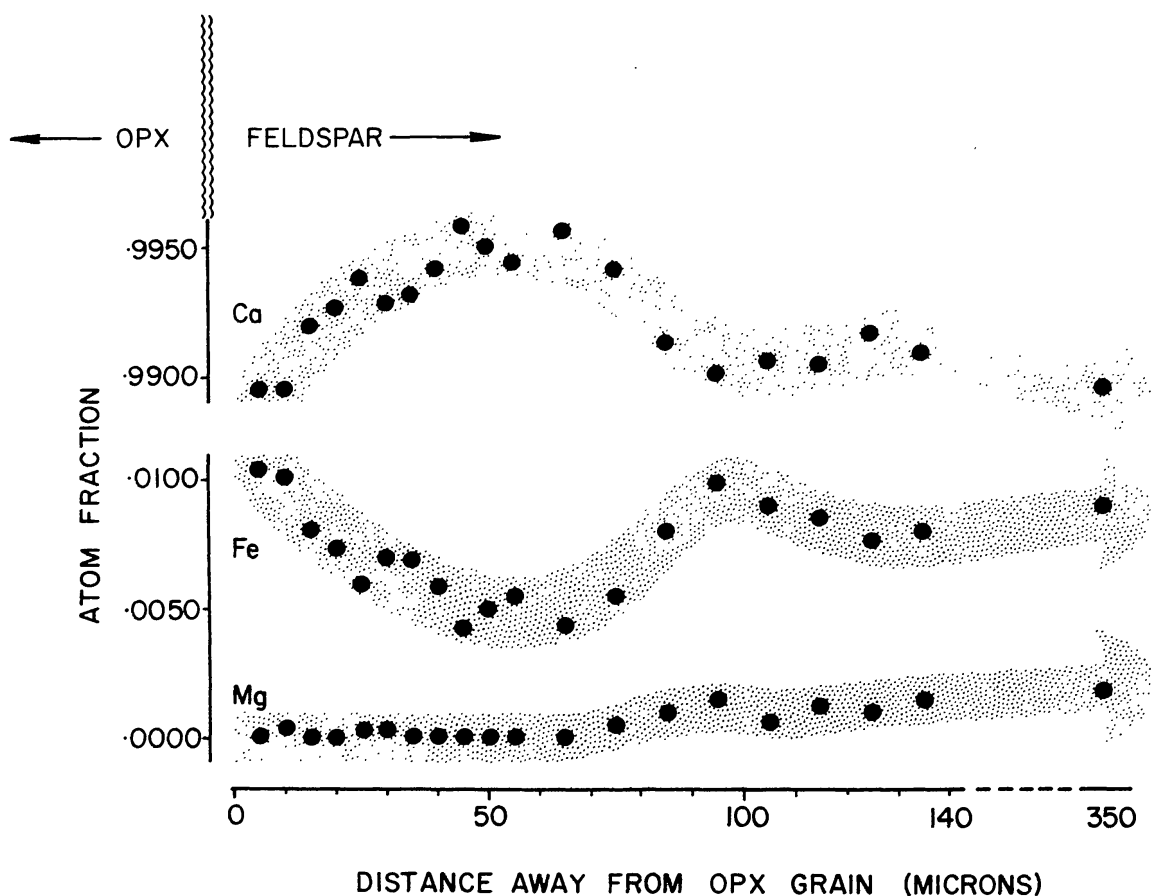


Fig. 7 (b) Diffusion profile: Ca, Fe, and Mg concentrations in feldspar versus distance from coexisting opx. Shaded band is same as Fig. 7a.

than 800°C. Fe–Mg intercrystalline exchange between augite and opx is expressed:

$$K_D = \frac{X_{\text{Fe(cpx)}}X_{\text{Mg(opx)}}}{X_{\text{Fe(opx)}}X_{\text{Mg(cpx)}}$$

Table 1 lists calculated K_D 's for the anorthosites reported here. Most fall below the average of 0.69 determined by Lindsley *et al.* (1974) for 810°C.

Sample 66043,1,4 has a K_D greater than 0.69 and the pyroxene equilibration has resulted in an augite less calcic and a hypersthene more calcic than those from Lindsley *et al.*'s (1974) 810°C isotherm. This anorthositic fragment is, therefore, believed to have equilibrated at substantially higher temperature than the other anorthosites, perhaps during a subsequent impact event.

Metamorphism was profound enough for intergranular cation diffusion to occur between feldspar and pyroxene (see Fig. 7). We believe that these cation diffusion trends reflect changes in the thermal environment.

The period of intense metamorphism hypothesized above may be explained either by slow cooling after initial lunar differentiation or by subsequent reheating by the intense moon-wide impact events at approximately 4 G.y. ago. We favor

Table 6. Selected microprobe analyses.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
SiO ₂	44.2	44.2	44.7	43.8	—	—	37.9	0.13	0.75	43.9	44.8	51.4	43.4	51.9	44.1	44.8
Al ₂ O ₃	26.3	26.6	26.6	35.5	0.02	0.17	—	17.4	14.3	35.5	35.6	0.46	35.2	0.77	35.7	34.5
TiO ₂	0.37	0.42	0.92	—	52.7	54.2	—	3.42	3.19	—	—	0.28	—	0.21	—	—
FeO	5.7	5.5	6.2	0.16	43.6	43.5	20.4	32.8	33.2	0.04	0.18	27.2	0.11	13.4	0.13	0.08
MnO	0.12	0.04	0.08	—	0.41	0.60	—	—	—	—	—	0.63	—	0.31	—	—
MgO	9.0	7.2	6.4	0.00	1.61	1.82	40.5	4.0	5.1	0.02	0.05	17.9	0.00	17.8	0.10	0.00
CaO	14.0	15.5	15.2	19.9	—	—	0.45	—	—	19.3	19.3	1.13	19.8	12.7	18.5	20.1
Na ₂ O	0.39	0.36	0.43	0.19	—	—	—	—	—	0.40	0.42	0.00	0.37	0.00	0.48	0.35
K ₂ O	0.08	0.09	0.19	0.00	—	—	—	—	—	0.00	0.00	0.09	0.06	—	0.00	0.02
Cr ₂ O ₃	0.09	0.13	—	—	0.08	0.16	—	43.7	42.4	—	—	—	—	0.21	—	—
Σ	100.3	100.0	100.7	99.6	98.4	100.5	99.3	101.5	98.9	99.2	100.4	99.1	98.9	97.9	99.0	99.9
Mg				—			78.0	—	—	—	—	52.6	—	51.2	—	—
Fe				—			22.0	—	—	—	—	45.0	—	22.6	—	—
Ca				98.3			—	—	—	96.4	96.2	2.4	96.4	26.2	95.5	96.8
Na				1.7			—	—	—	3.6	3.8	—	3.2	—	4.5	3.1
K				0.0			—	—	—	0.0	0.0	—	0.4	—	0.0	0.1
(1 and 2) 60015,113 Splashed glass.																
(3) 50052 Soil Average of glass (Simkin <i>et al.</i> , 1973).																
(4) 60015,113 Feldspar clast in splashed glass.																
(5 and 6) 60025,117 Ilmenite.																
(7) 60215,13 Olivine clast.																
(8) 60215,13 Chrome spinel in large anorthosite clast.																
(9) 61016,217 Chrome spinel.																
(10) 63503,13,12 Feldspar.																
(11) 64503,1,26 Feldspar.																
(12) 64503,1,36 Orthopyroxene.																
(13) 64503,1,36 Feldspar.																
(14) 65315,89 Broad beam analysis-exsolved pyroxene.																
(15) 66043,1,4 Feldspar.																
(16) 15415,95 Feldspar.																

the slow cooling hypothesis. The probable depths of accumulation of these anorthosites seems an excellent setting for slow cooling, and the solid-state pyroxene migration in the anhydrous and low-temperature conditions evident must certainly had occurred at extremely slow rates. Thus, a deep-seated region of extremely prolonged, undisturbed metamorphism seems likely. Stewart (1975) has termed this metamorphism as "Apollonian."

The excavation of the large anorthosite fragments probably occurred prior to 4 G.y. after which the meteorite flux showed a marked decline. ^{40}Ar – ^{39}Ar ages for such rocks as 15415, 60025, and 67075 cluster around this date. The impact events which were energetic enough to create the mare basins would also be expected to have sampled the anorthosite cumulate and imposed the primary cataclastic textures. These impacts, from purely geometric considerations, must certainly have sampled a greater volume of shallow anorthosite than deeper anorthosite. If our simple lithologic model is correct, then we would expect to see a sampling bias toward the shallower or Mg-rich pyroxene-bearing anorthosites. Such a relationship is observed in the thirteen anorthosites studied here (see Fig. 2). The data of Dowty *et al.* (1974) presents a similar bias towards Mg-rich compositions in their suite of eleven anorthositic rake samples. These observations add support to the hypothesis of Fe enrichment with depth in the anorthosite layer.

Krähenbühl *et al.* (1973) suggest on the basis of the low-meteoritic component in Apollo 16 anorthosites that the anorthosites had a deep-seated origin. They point to Nectaris as a likely source due to its probable depth and its proximity to the Apollo 16 sampling sites. We agree with this possible origin but also present the possibility of a more local source. In a discussion of the stratigraphy of the Apollo 16 site, Head (1975) points out that a 150-km crater ("Unnamed Crater A") which encloses the landing site is the oldest crater in that region and may, in fact, predate Nectaris. Such a large crater is likely to have sampled to depths of tens of kilometers.

The subsequent surface and near-surface history of the Apollo 16 anorthosites involved impact shock, local melting, and brecciation. Type II anorthosites witnessed shock-induced heating to temperatures high enough to initiate partial melting. The 3.5-G.y. age for 60015,12 (Schaeffer and Husain, 1974) probably attests to such an event. The brecciation of rocks such as 60215, 65315, and 67075 seems likely to have occurred at shallow depths (perhaps under a blanket of impact ejecta). Mixing of different highland lithologic types resulted in formation of some anorthosite-rich breccias. Smaller scale crater-forming events eventually brought all the samples to the surface where they were recovered by the Apollo 16 astronauts.

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